ELE 420/520 Cyber-Physical System Security

Project 4 **Due: April 24th, 11:59 pm EST**

Use Zeek to Detect Man-in-the-Middle Attacks

1 Preparation

In Project 3, we assumed that the data aggregator was compromised by attackers. The attackers use the data aggregator to perform a man-in-the-middle attack: modify measurements when the data aggregator combines measurements received from relays. For this project, I have added a DNP3m analyzer that can be loaded by Zeek to extract all measurements from the network communications. Since the DNP3m analyzer can extract the measurements from the packets observed in the network, we can write Zeek scripts to process or analyze the measurements.

2 Implementation

In this project, the assignment is to use Zeek and its script language to detect the man-in-the-middle attack. The high-level detection logic is as follows: using the source/destination IP addresses to identify the measurements sent from the relay to the Data Aggregators and the measurements sent from the Data Aggregator to the Control Center. Then in the script, you can compare the measurements from the same index to determine whether the measurements are modified by the Data Aggregator or not. If there are modifications, you can print them out. Please download the starter code at https://github.com/hugolin615/ELE420520Spring2026/tree/main/project4.

Specifically, your implementation is marked by TODO in the following files:

- ***detect_mitm.zeek***: rely on the aforementioned detection logic to implement this Zeek script. This is the script that you will load with Zeek.

You should run Zeek in two following scenarios:

- ***Run Zeek to analyze a network trace (a pcap file).*** You can use the following command "***zeek -Cr data_aggregator.pcap ./dnp3m_analyzer/dnp3m-analyzer.hlto ./detect_mitm.zeek***" under the project directory. If your script is working properly, Zeek will create a log file named "dnp3m.log", which simply prints out the measurements that the Data Aggregator received from relay machines and the measurements that the Data Aggregator delivered to the Control Center. These two sets of measurements should be consistent with each other, as we use Zeek to analyze the *data_aggregator.pcap*, which contains no attacks; you should not observe any messages with ALERTs. Rename this log file to "no_mitm.log".

- ***Run Zeek to analyze runtime network communication.*** After opening Xterm consoles for Control Center, Data Aggregators, Relay1, Relay2, Relay3, and Relay4, you should open another Xterm console for the Data Aggregator and run Zeek by using the following command "***zeek -C -i da-eth0 ./dnp3m_analyzer/dnp3m-analyzer.hlto ./detect_mitm.zeek***". Then it would be best if you ran the Python scripts of Control Center, Data Aggregators, Relay1, Relay2, Relay3, and Relay4 to perform the experiments in Project 3. It is critical to run Zeek before you run Python scripts to perform experiments. Otherwise, Zeek may miss some network packets, and your scripts may not work properly. After the experiment ends, you can terminate Zeek by typing "Control C" in the Xterm console where you run Zeek. The Zeek will create a log file named "dnp3m.log". If your script is working properly, dnp3m.log should include ALERT messages in addition to the printout measurements. Rename this log file with the name "mitm.log".

3 Turn-in and Grading.

Please submit a single PDF file, named as "*Last Name_First Name_Project4.pdf*", including the following contents:

- **Zeek scripts, i.e., detect_mitm.zeek (45%).**
- **Logs files (no_mitm.log and mitm.log) (30%)**.
- **A snapshot of running the Python scripts in Mininet (25%).** After running the scripts, take a snapshot to show the consoles of all machines. I suggested putting machines in the pattern shown in Figure 1.

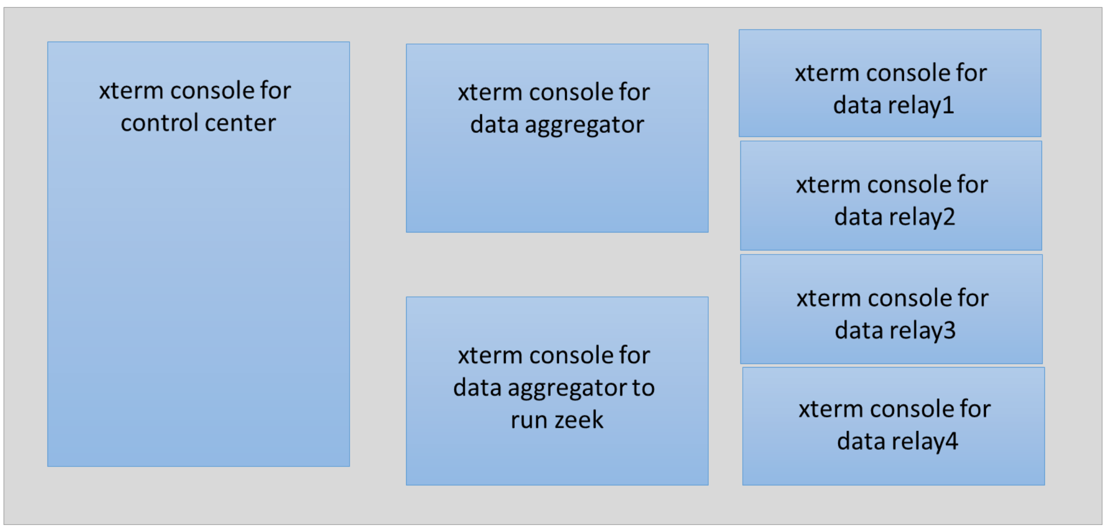

Figure 1. Suggested pattern to show all running machines in Mininet.